Lab 5 - Linux Firewall Exploration - Guide & Report

Course: NT140 - Network Security
Student: [Your Name] - [MSSV]
Date: [Date]

Network Topology

```
┌──────────────────┐
│   VM B           │ IP: 192.168.35.145
│   (Server)       │ (NAT/VMnet8)
└──────────────────┘
         │
         │
         │ 192.168.35.0/24
         │
    ┌────────────────┐
    │  pfSense       │ WAN: 192.168.35.146
    │  (Firewall)    │ LAN: 192.168.132.250
    └────────────────┘
         │
         │
         │ 192.168.132.0/24
         │
┌──────────────────┐
│   VM A           │ IP: 192.168.132.101
│   (Client)       │ (Host-only/VMnet2)
└──────────────────┘
```

Network Summary:

- **WAN (VMnet8/NAT):** 192.168.35.0/24
  - pfSense WAN: 192.168.35.146
  - VM B: 192.168.35.145
- **LAN (VMnet2/Host-only):** 192.168.132.0/24
  - pfSense LAN: 192.168.132.250
  - VM A: 192.168.132.101

PART 1: Create pfSense VM

Step 1: Create VM in VMware

→ **ON HOST:**

```
vmware &
```

1. File → New Virtual Machine
2. Typical → Next
3. Installer disc image: Select pfSense ISO → Next
4. Guest OS: Other → FreeBSD 12 64-bit → Next
5. Name: pfSense-Firewall → Next
6. Disk: 20 GB, single file → Next
7. Customize Hardware:
   - **Memory**: 2048 MB
   - **Processors**: 2 cores
   - **Network Adapter**: NAT (this is WAN/em0)
   - Click **Add** → Network Adapter → Custom VMnet2 (this is LAN/em1)
   - **CD/DVD**: pfSense ISO connected
8. Close → Finish

Verify: pfSense VM has 2 network adapters

Step 2: Install pfSense

→ **START pfSense VM**

→ **INSIDE pfSense CONSOLE:**

1. Wait for boot (auto-boot in 10s or press Enter)
2. Accept copyright → Enter
3. Select **Install** → Enter

[📸 Screenshot: Install menu selected]

4. Keymap: **Continue with default keymap** → Enter
5. Partitioning: **Auto (UFS) BIOS** → Enter
6. **Entire Disk** → Enter
7. Confirm **Yes** → Enter
8. **Wait 5-10 minutes** for installation

[📸 Screenshot: Installation in progress]

9. Select **No** (don't open shell) → Enter
10. Select **Reboot** → Enter

Step 3: Remove ISO

→ AFTER REBOOT STARTS, SWITCH TO HOST:

```
# Stop pfSense VM
vmrun stop ~/vmware/pfSense-Firewall/pfSense-Firewall.vmx

# Edit VMX file
nano ~/vmware/pfSense-Firewall/pfSense-Firewall.vmx

# Find line with ISO (ide1:0 or sata):
# ide1:0.fileName = "/path/to/pfSense.iso"

# Change to:
# ide1:0.fileName = "auto detect"

# Save: Ctrl+O, Enter, Ctrl+X

# Start pfSense VM again
vmrun start ~/vmware/pfSense-Firewall/pfSense-Firewall.vmx
```

PART 2: Configure pfSense Interfaces

→ INSIDE pfSense CONSOLE (after reboot):

pfSense boots and detects network interfaces.

[📸 Screenshot: pfSense detecting em0 and em1]

Assign Interfaces

```
Should VLANs be set up now [y|n]?
→ n
→ Enter

Enter the WAN interface name or 'a' for auto-detection:
→ em0
→ Enter

Enter the LAN interface name or 'a' for auto-detection:
```

```
→ em1
→ Enter

Enter Optional interface name (press Enter if done):
→ Enter

Do you want to proceed [y|n]?
→ y
→ Enter
```

Wait 20-30 seconds for configuration

```
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 46.850/47.233/47.654/0.329 ms

Press ENTER to continue.

VMware Virtual Machine - Netgate Device ID: 2a355706b103b46126a0

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

 WAN (wan)       -> em0        -> v4/DHCP4: 192.168.35.146/24
 LAN (lan)       -> em1        -> v4: 192.168.132.250/24

 0) Logout (SSH only)                  9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart webConfigurator
 3) Reset webConfigurator password    12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option:
```

PART 3: Configure VM A (Client)

Step 1: Change Network to VMnet2

→ ON HOST:

```
# Stop VM A
vmrun stop ~/vmware/VM-A/VM-A.vmx

# Edit VMX
nano ~/vmware/VM-A/VM-A.vmx

# Find ethernet0 section, change to:
ethernet0.connectionType = "custom"
```

```
ethernet0.vnet = "vmnet2"

# Save and exit

# Start VM A
vmrun start ~/vmware/VM-A/VM-A.vmx
```

Step 2: Get IP from pfSense DHCP

→ INSIDE VM A (Ubuntu):

```
# Check interface name
ip link show

# Assume interface is ens33
# Request DHCP
sudo dhclient -v ens33

# Or restart network
sudo systemctl restart NetworkManager

# Wait 10-20 seconds, then check IP
ip addr show ens33

# Should show: inet 192.168.132.101/24 (or 100-200 range)
```

```
       valid_lft forever preferred_lft forever
undertaker@wekone06:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 00:0c:29:19:8f:a9 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
undertaker@wekone06:~$ ip addr show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:19:8f:a9 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.132.101/24 metric 1024 brd 192.168.132.255 scope global dynamic ens33
       valid_lft 2724sec preferred_lft 2724sec
    inet6 fe80::36dd:1648:49f2:4e5e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
undertaker@wekone06:~$
```

```
# Verify gateway
ip route show | grep default

# Should show: default via 192.168.132.250 dev ens33
```

```
undertaker@wekone06:~$ ip route show | grep default
default via 192.168.132.1 dev ens33 proto dhcp src 192.168.132.101 metric 1024
```

Step 3: Test Connectivity to pfSense

→ INSIDE VM A:

```
# Ping pfSense
ping -c 4 192.168.132.250

# Should see: 0% packet loss
```

```
undertaker@wekone06:~$ ping -c 4 192.168.132.250
PING 192.168.132.250 (192.168.132.250) 56(84) bytes of data.
64 bytes from 192.168.132.250: icmp_seq=1 ttl=64 time=1.83 ms
64 bytes from 192.168.132.250: icmp_seq=2 ttl=64 time=0.361 ms
64 bytes from 192.168.132.250: icmp_seq=3 ttl=64 time=0.544 ms
64 bytes from 192.168.132.250: icmp_seq=4 ttl=64 time=0.434 ms

--- 192.168.132.250 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3029ms
rtt min/avg/max/mdev = 0.361/0.793/1.833/0.603 ms
undertaker@wekone06:~$
```

```
# Test web interface port
curl -I http://192.168.132.250

# Should return HTTP response
```

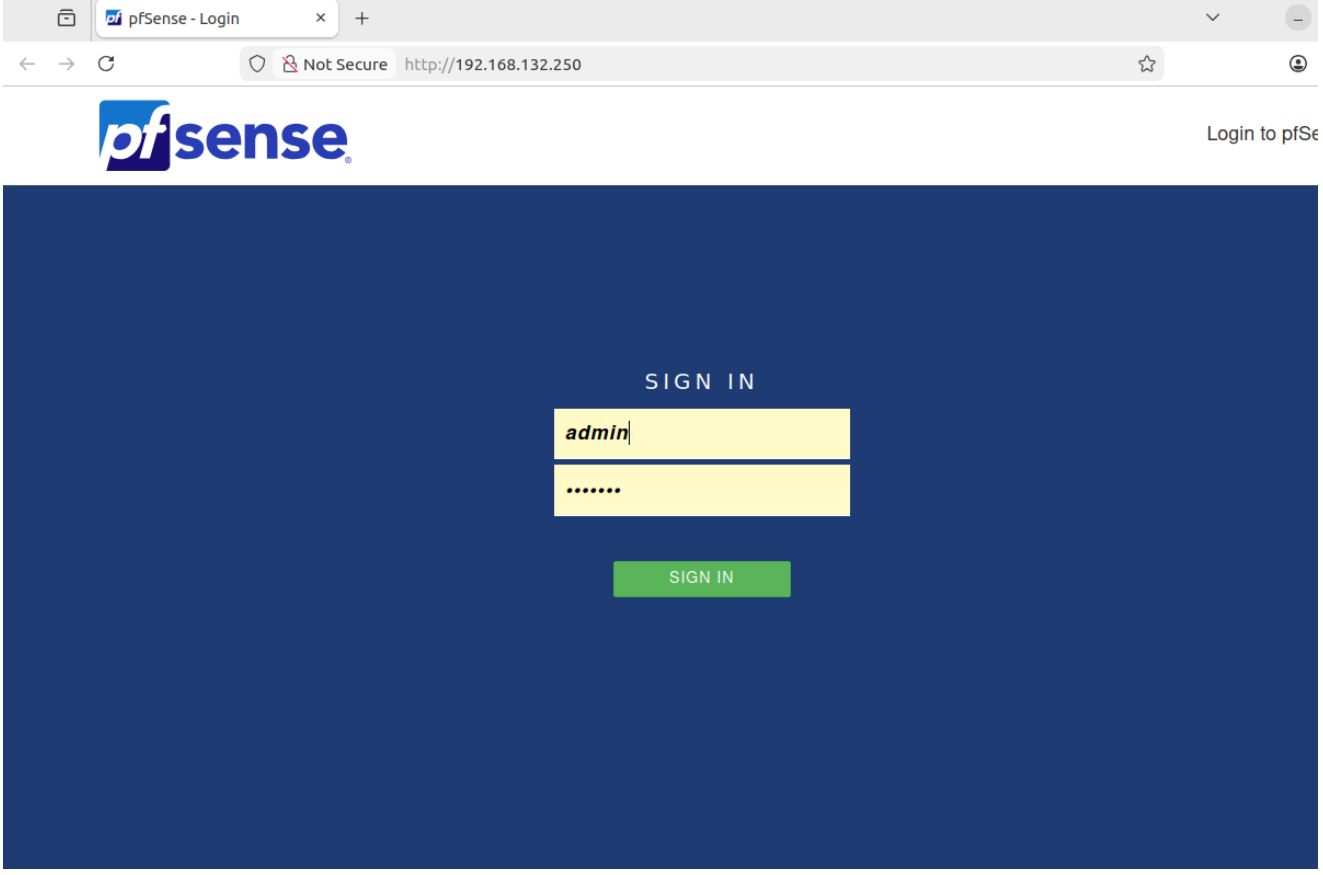
```
undertaker@wekone06:~$ curl -I http://192.168.132.250
HTTP/1.1 200 OK
Server: nginx
Date: Thu, 15 Jan 2026 10:47:27 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive
X-Frame-Options: SAMEORIGIN
Last-Modified: Thu, 15 Jan 2026 10:47:27 GMT
Set-Cookie: PHPSESSID=0464eaa76cd92a36568ab55cbf2a9a4a; path=/; HttpOnly
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
X-Frame-Options: SAMEORIGIN

undertaker@wekone06:~$
```

PART 4: Access pfSense WebUI

→ INSIDE VM A:

```
# Launch Firefox
firefox http://192.168.132.250 &
```

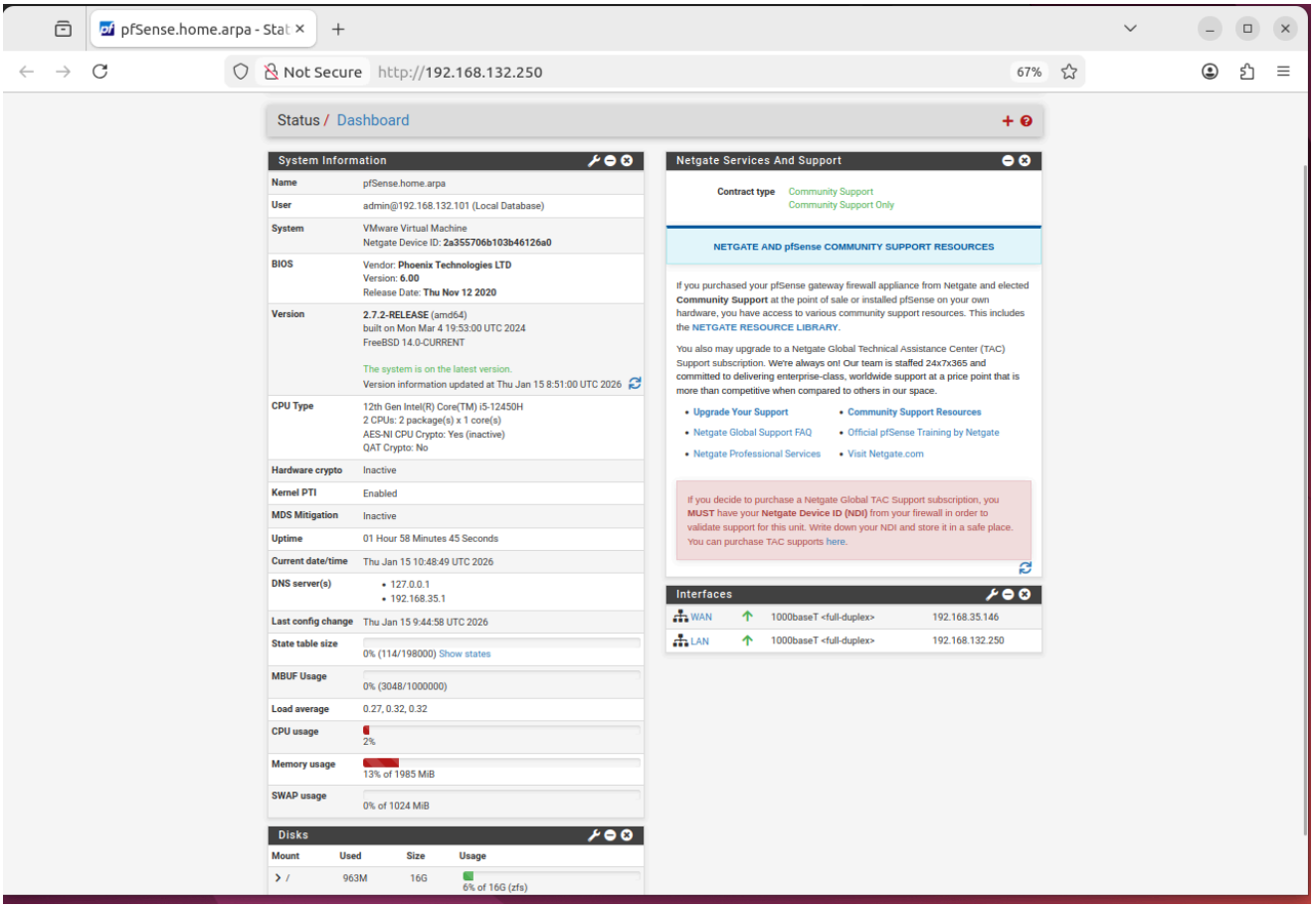
Login page appears



Login:

- Username: `admin`
- Password: `pfsense`

Click **Sign In**



PART 5: Configure VM B (Server)

Step 1: Set Static IP on VM B

→ INSIDE VM B (Ubuntu):

```
# Check interface
ip link show

# Assume ens33
# Set static IP (temporary)
sudo ip addr flush dev ens33
sudo ip addr add 192.168.35.145/24 dev ens33
sudo ip route add default via 192.168.35.2
sudo ip link set ens33 up

# Verify
ip addr show ens33
```

For permanent (netplan):

```
sudo nano /etc/netplan/01-netcfg.yaml
```

Add:

```
network:
  version: 2
  ethernets:
    ens33:
      addresses:
        - 192.168.35.145/24
      gateway4: 192.168.35.2
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

```
# Apply
sudo netplan apply

# Verify
ip addr show ens33
ping -c 4 8.8.8.8
```

[📸 Screenshot: VM B showing IP 192.168.35.145]

Step 2: Install Telnet & SSH Servers

→ **INSIDE VM B:**

```
# Update packages
sudo apt update

# Install xinetd and telnet server
sudo apt install -y xinetd telnetd

# Create telnet configuration
echo 'service telnet
{
    disable = no
    flags = REUSE
    socket_type = stream
    wait = no
    user = root
    server = /usr/sbin/in.telnetd
    log_on_failure += USERID
}' | sudo tee /etc/xinetd.d/telnet

# Restart xinetd
sudo systemctl restart xinetd
sudo systemctl enable xinetd

# Check xinetd status
sudo systemctl status xinetd
```

```
# Install SSH server
sudo apt install -y openssh-server

# Start SSH
sudo systemctl start ssh
sudo systemctl enable ssh

# Check SSH status
sudo systemctl status ssh
```

```
# Verify both services running
sudo ss -tuln | grep -E ':23|:22'

# Should show:
# tcp   LISTEN   0   64   0.0.0.0:23   0.0.0.0:*
# tcp   LISTEN   0   128  0.0.0.0:22   0.0.0.0:*
```

[📸 Screenshot: VM B showing telnet (23) and SSH (22) listening ports]

TASK 1: Firewall Rules Configuration

Task 1.1: Block Ping to VM B

Objective: Block ping from VM A (LAN) to VM B (192.168.35.145)

→ **INSIDE VM A Browser (pfSense WebUI):**

Step 1: Navigate to Rules

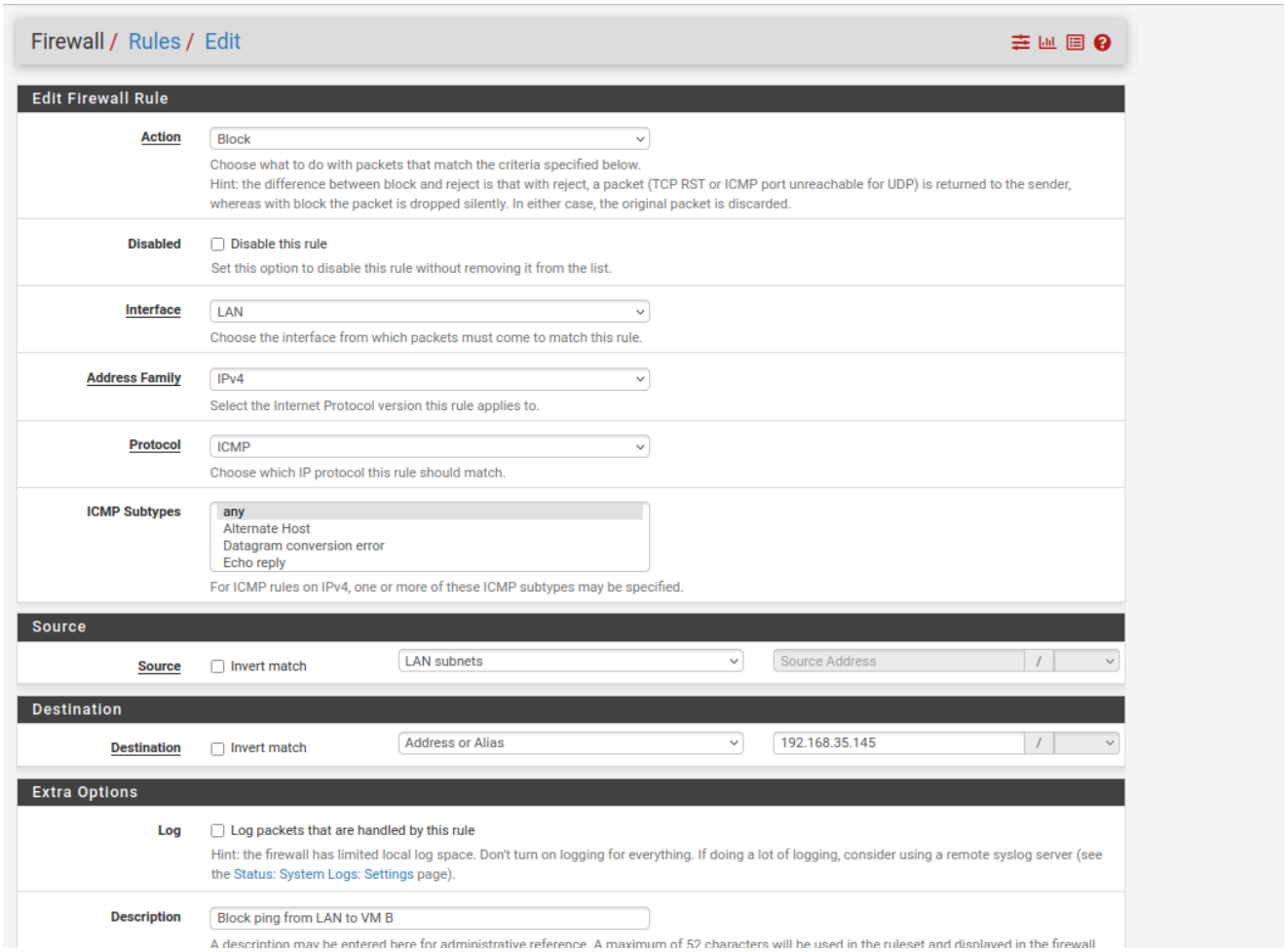
1. Click **Firewall** menu
2. Click **Rules**
3. Click **LAN** tab
4. Click ↑ **Add** button (up arrow = add to top)

Step 2: Configure Block Rule
Rule 1:

Action: Pass
Interface: LAN
Protocol: any
Source: LAN net
Destination: any
Description: Allow LAN to any (default)

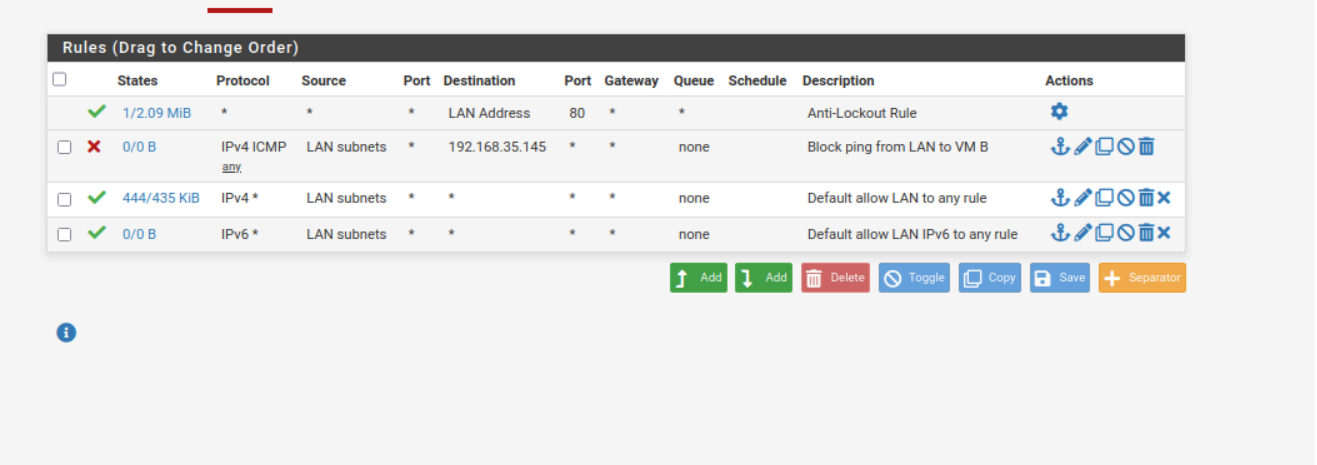
Rule 2:
Fill in these fields:

- **Action**: Block
- **Disabled**: (leave unchecked)
- **Interface**: LAN
- **Address Family**: IPv4
- **Protocol**: ICMP
- **ICMP Subtype**: any
- **Source**: LAN net
- **Destination**: Single host or alias
  - Type: `192.168.35.145`
- **Description**: Block ping from LAN to VM B



Step 3: Save and Apply

1. Scroll down → Click **Save**
2. Click **Apply Changes**



Verification:

→ INSIDE VM A Terminal:

Test 1: Ping VM B (should FAIL):

```
ping -c 5 192.168.35.145

# Expected result:
```

```
# Request timeout OR 100% packet loss
# No response from 192.168.35.145
```

```
undertaker@wekone06:~$ ping -c 5 192.168.35.145
PING 192.168.35.145 (192.168.35.145) 56(84) bytes of data.

--- 192.168.35.145 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4092ms
```

Test 2: Ping pfSense (should SUCCESS):

```
ping -c 4 192.168.132.250

# Expected result:
# 64 bytes from 192.168.132.250...
# 0% packet loss
```

```
undertaker@wekone06:~$ ping -c 4 192.168.132.250
PING 192.168.132.250 (192.168.132.250) 56(84) bytes of data.
64 bytes from 192.168.132.250: icmp_seq=1 ttl=64 time=0.432 ms
64 bytes from 192.168.132.250: icmp_seq=2 ttl=64 time=0.694 ms
64 bytes from 192.168.132.250: icmp_seq=3 ttl=64 time=0.474 ms
64 bytes from 192.168.132.250: icmp_seq=4 ttl=64 time=0.582 ms

--- 192.168.132.250 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3085ms
rtt min/avg/max/mdev = 0.432/0.545/0.694/0.101 ms
```

Test 3: Ping Internet (should SUCCESS):

```
ping -c 4 8.8.8.8

# Expected result: 0% packet loss
```

```
undertaker@wekone06:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=127 time=88.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=127 time=46.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=127 time=49.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=127 time=48.1 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 46.410/57.975/88.150/17.450 ms
undertaker@wekone06:~$
```

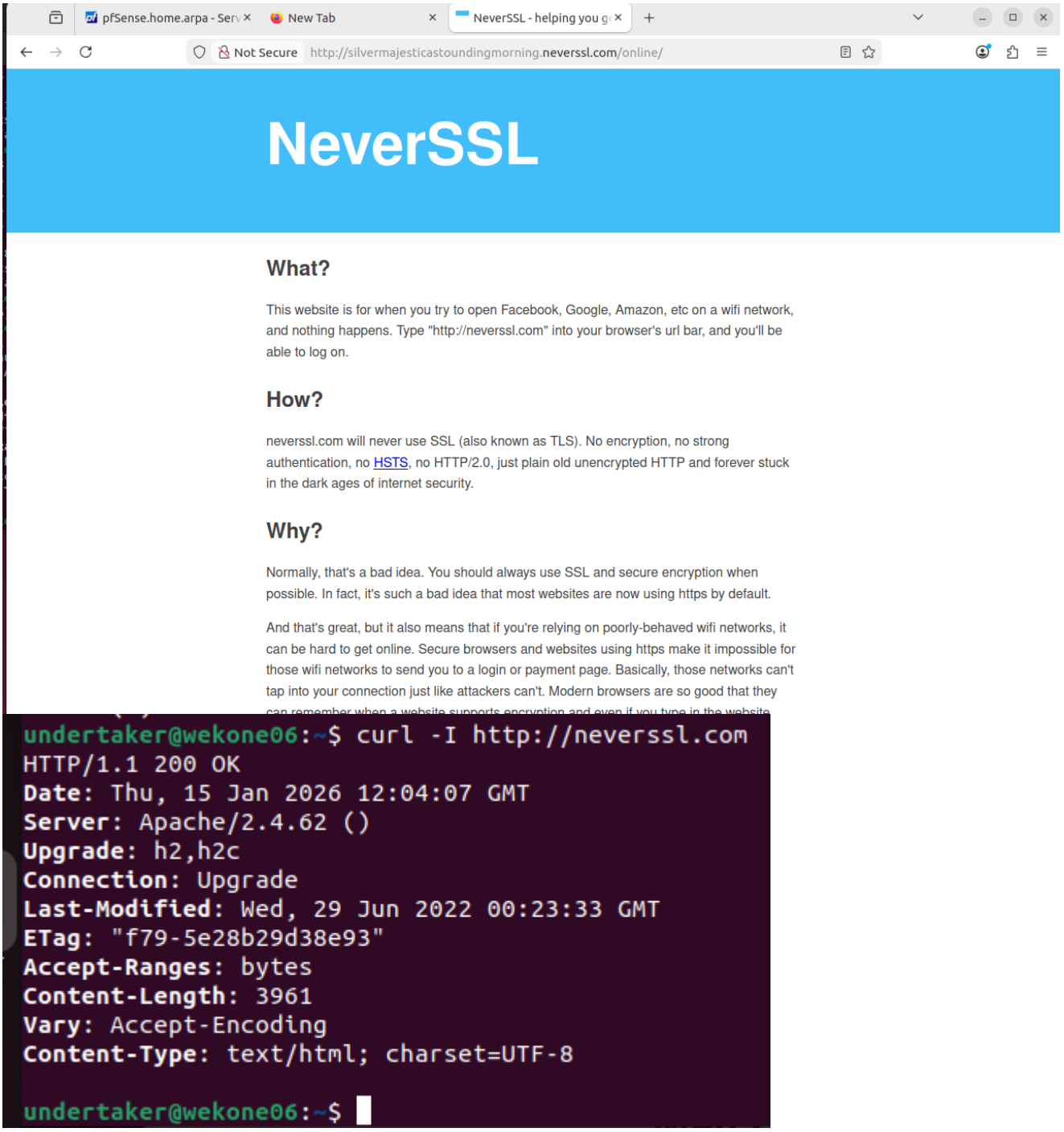
Task 1.2: Block HTTP Traffic

Objective: Block HTTP (port 80) from LAN to any destination

Before blocking - Test HTTP access:

→ INSIDE VM A Browser:

- Open: `http://neverssl.com`



→ INSIDE VM A Terminal:

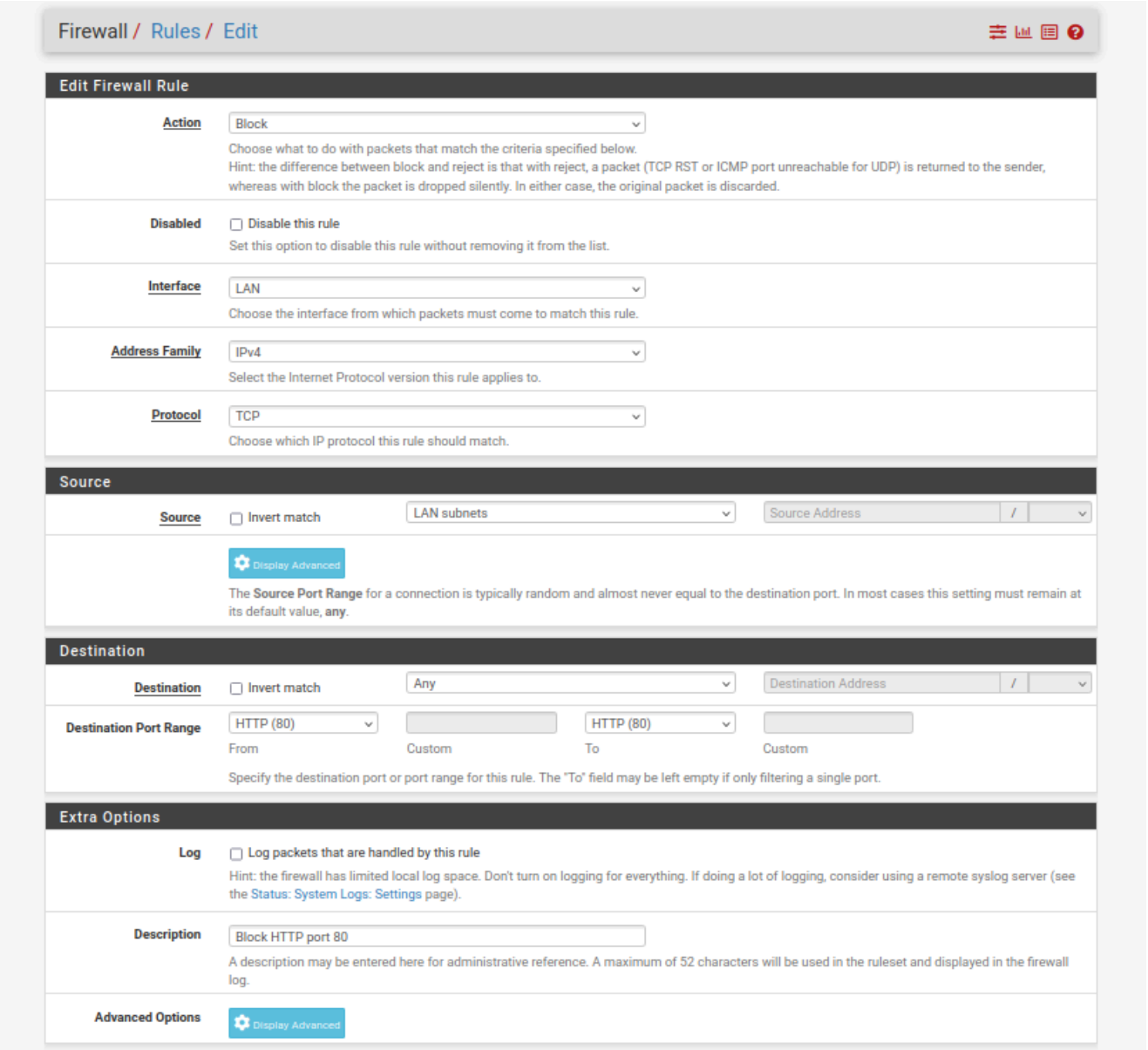
```
curl -I http://neverssl.com
# Should return: HTTP/1.1 200 OK
```

Configure Block Rule:

→ **INSIDE VM A Browser (pfSense):**

1. Firewall → Rules → LAN → **Add**
2. Configure:
   - **Action**: Block
   - **Interface**: LAN
   - **Address Family**: IPv4
   - **Protocol**: TCP
   - **Source**: LAN net
   - **Destination**: any
   - **Destination Port Range**:
     - From: HTTP (80)
     - To: HTTP (80)
   - **Description**: Block HTTP port 80
3. **Save** → **Apply Changes**

**



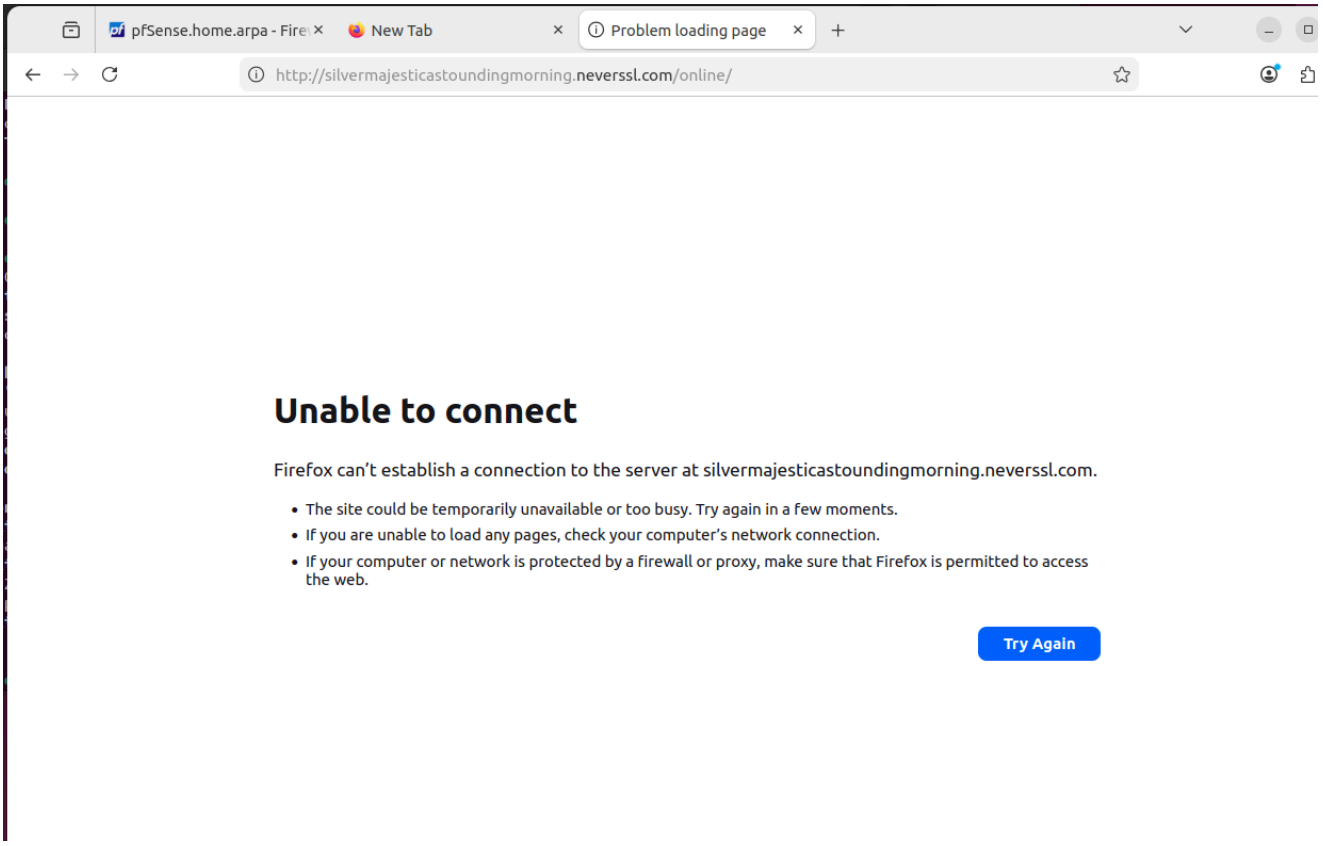
Verification:

→ INSIDE VM A Terminal:

```
# Test HTTP (should FAIL)
curl -I http://neverssl.com
# Expected: Connection timeout or refused

curl -I http://example.com
# Expected: Connection timeout

# Test HTTPS (should SUCCESS)
curl -I https://www.google.com
# Expected: HTTP/2 200 (HTTPS still works)
```

→ INSIDE VM A Browser:

- Try: `http://neverssl.com`



- Try: `https://www.google.com`
- Should work (HTTPS not blocked)

```
undertaker@wekone06:~$ curl -I http://neverssl.com
^C
undertaker@wekone06:~$ curl -I http://example.com
^C
undertaker@wekone06:~$ curl -I https://www.google.com
HTTP/2 200
content-type: text/html; charset=ISO-8859-1
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-Kx2kKqvrEK2KhW2x8RF2tg' 'stric
t-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/oth
er-hp
accept-ch: Sec-CH-Prefers-Color-Scheme
p3p: CP="This is not a P3P policy! See g.co/p3phelp for more info."
date: Thu, 15 Jan 2026 12:09:01 GMT
server: gws
x-xss-protection: 0
x-frame-options: SAMEORIGIN
expires: Thu, 15 Jan 2026 12:09:01 GMT
cache-control: private
set-cookie: AEC=AaJma5tJGbJ-JaZmu1Y5fXor2m5YGBL42q9x6UUtrNWCp94JZShodIFAciE; expires=Tue, 14-Jul-2026 12:09:01 GMT; pat
h=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
set-cookie: NID=528=IgeOnOCfOxU-ecjEnkyvJ1-3ch_7p1kr4FhSJU5bqHpcsELyYW1i4_-jLg0PtgJU97atAkw_Tb93KUbq_Ku1Vl8MZKYloKcmnVD
jveO2WnzZcilwhLP7_W0ErAjO0nM8K4A1axutYMzXOlSxJiCc1hY1bXFU1zkOAaawMNANQGqXAcQ-AppSUp4vRgA_U1hPJPQkPt3FagroSGHzVuvLSMiMVE
03YQ; expires=Fri, 17-Jul-2026 12:09:01 GMT; path=/; domain=.google.com; HttpOnly
set-cookie: __Secure-BUCKET=CO4B; expires=Tue, 14-Jul-2026 12:09:01 GMT; path=/; domain=.google.com; Secure; HttpOnly
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

undertaker@wekone06:~$
```

Result: ✅ HTTP blocked, HTTPS works

Task 1.3: Block Telnet

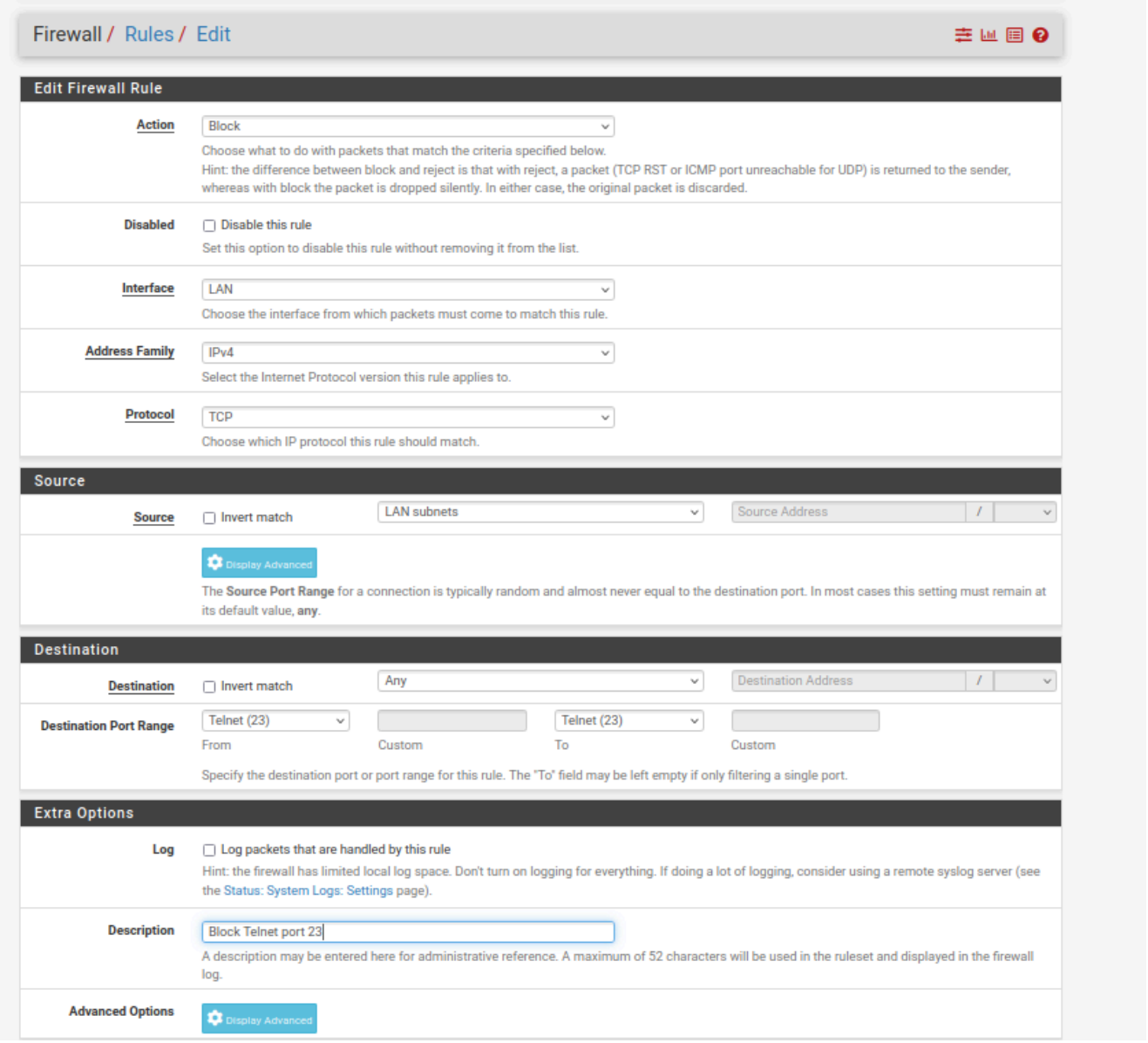
Objective: Block Telnet (port 23) from LAN to any destination

Configure Block Rule:

→ INSIDE VM A Browser (pfSense):

1. Firewall → Rules → LAN → **Add**
2. Configure:
   - **Action**: Block
   - **Interface**: LAN
   - **Address Family**: IPv4
   - **Protocol**: TCP
   - **Source**: LAN net
   - **Destination**: any
   - **Destination Port Range**:
     - From: 23
     - To: 23
   - **Description**: Block Telnet port 23
3. **Save → Apply Changes**

**



Verification:

→ **INSIDE VM A Terminal:**

```
# Test Telnet (should FAIL)
telnet 192.168.35.145

# Expected: Connection timeout or refused

# Test with timeout
timeout 10 telnet 192.168.35.145

# Test with netcat
nc -zv -w 5 192.168.35.145 23
# Expected: Connection refused or timeout

# Test SSH (should SUCCESS)
nc -zv 192.168.35.145 22
# Expected: Connection succeeded

ssh user@192.168.35.145
# Should work (SSH not blocked)
```

```
undertaker@wekone06:~$ telnet 192.168.35.145
Trying 192.168.35.145...
^C
undertaker@wekone06:~$ timeout 10 telnet 192.168.35.145
Trying 192.168.35.145...
undertaker@wekone06:~$ nc -zv -w 5 192.168.35.145 23
nc: connect to 192.168.35.145 port 23 (tcp) timed out: Operation now in progress
undertaker@wekone06:~$ nc -zv 192.168.35.145 22
Connection to 192.168.35.145 22 port [tcp/ssh] succeeded!
undertaker@wekone06:~$ ssh user@192.168.35.145
The authenticity of host '192.168.35.145 (192.168.35.145)' can't be established.
ED25519 key fingerprint is SHA256:jFmib9DNLnUafTq2FdUEzzWprlUF3fGNW9ihR0ywvx8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.35.145' (ED25519) to the list of known hosts.
user@192.168.35.145's password:
Permission denied, please try again.
user@192.168.35.145's password:
Permission denied, please try again.
user@192.168.35.145's password:
user@192.168.35.145: Permission denied (publickey,password).
undertaker@wekone06:~$
```

Result: ✅ Telnet blocked, SSH works

Task 1.4: Block YouTube & Facebook

Objective: Block YouTube and Facebook using DNS Override

Why DNS Override?

- YouTube/Facebook use CDNs with 100+ different IPs
- IPs change constantly
- Blocking by IP = impossible
- DNS Override = return fake IP (0.0.0.0)

Configure DNS Override:

→ **INSIDE VM A Browser (pfSense):**

1. Navigate to: **Services** → **DNS Resolver**
2. Scroll down to **Host Overrides** section
3. Click **Add** button

Add Override #1:

- Host: `www`
- Domain: `youtube.com`
- IP address: `0.0.0.0`
- Description: `Block YouTube`
- Click **Save**

Add Override #2:

- Host: `*` (asterisk = wildcard)
- Domain: `youtube.com`
- IP address: `0.0.0.0`
- Description: `Block all YouTube subdomains`
- Click **Save**

Add Override #3:

- Host: `*`
- Domain: `youtu.be`
- IP address: `0.0.0.0`
- Description: `Block YouTube short links`
- Click **Save**

Add Override #4:

- Host: `www`
- Domain: `facebook.com`
- IP address: `0.0.0.0`
- Description: `Block Facebook`

- Click **Save**

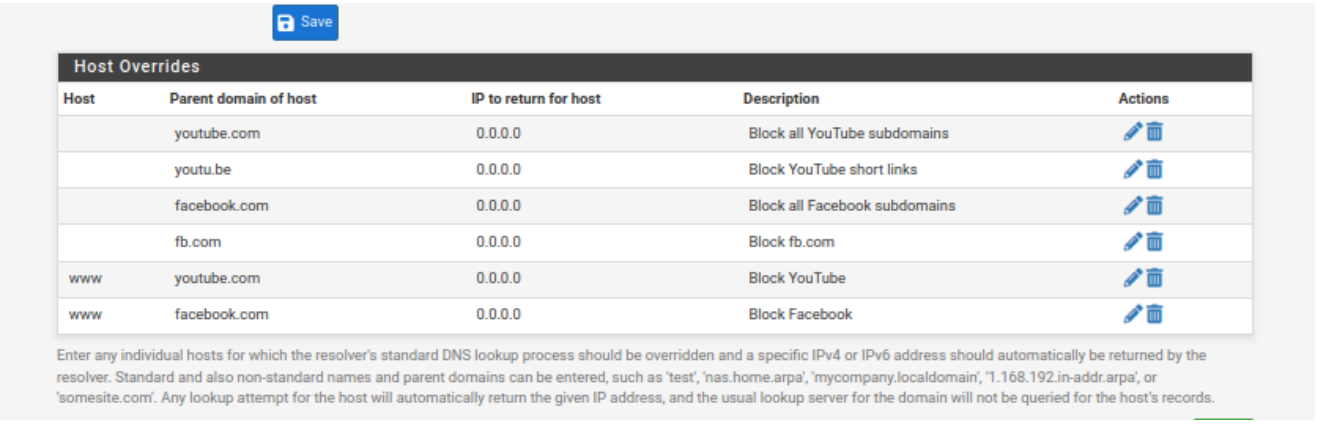
Add Override #5:

- Host: `*`
- Domain: `facebook.com`
- IP address: `0.0.0.0`
- Description: `Block all Facebook subdomains`
- Click **Save**

Add Override #6:

- Host: `*`
- Domain: `fb.com`
- IP address: `0.0.0.0`
- Description: `Block fb.com`
- Click **Save**

4. Click **Apply Changes** at top



Verification:

→ INSIDE VM A Terminal:

```
# Test YouTube DNS
nslookup www.youtube.com

# Expected output:
# Server:    127.0.0.53
# Address:   127.0.0.53#53
# Name:      www.youtube.com
# Address:   0.0.0.0

# Test Facebook DNS
nslookup www.facebook.com
```

```
# Expected output:
# Address:   0.0.0.0

# Using dig
dig www.youtube.com +short
# Expected: 0.0.0.0

dig www.facebook.com +short
# Expected: 0.0.0.0

# Test other sites (should work)
nslookup www.google.com
# Expected: Real IP addresses (like 142.250.x.x)
```

```
undertaker@wekone06:~$ nslookup www.youtube.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   www.youtube.com
Address: 0.0.0.0

undertaker@wekone06:~$ nslookup www.facebook.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   www.facebook.com
Address: 0.0.0.0

undertaker@wekone06:~$ dig www.youtube.com +short
0.0.0.0
undertaker@wekone06:~$ dig www.facebook.com +short
0.0.0.0
undertaker@wekone06:~$ nslookup www.google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   www.google.com
Address: 74.125.200.104
Name:   www.google.com
Address: 74.125.200.147
Name:   www.google.com
Address: 74.125.200.99
Name:   www.google.com
Address: 74.125.200.106
Name:   www.google.com
Address: 74.125.200.103
Name:   www.google.com
Address: 74.125.200.105
Name:   www.google.com
Address: 2404:6800:4003:c00::93
Name:   www.google.com
Address: 2404:6800:4003:c00::6a
Name:   www.google.com
Address: 2404:6800:4003:c00::67
Name:   www.google.com
Address: 2404:6800:4003:c00::68

undertaker@wekone06:~$ ===
```

→ **INSIDE VM A Browser:**

Try to access:

- `https://www.youtube.com`
- `https://youtube.com`
- `https://www.facebook.com`
- `https://facebook.com`

All should fail to load.

pfSense.home.arpa Problem loading pa Problem loading pa Problem loading pa Problem loading

www.youtube.com

Unable to connect

Firefox can't establish a connection to the server at www.youtube.com.

- The site could be temporarily unavailable or too busy. Try again in a few moments.
- If you are unable to load any pages, check your computer's network connection.
- If your computer or network is protected by a firewall or proxy, make sure that Firefox is permitted to acc the web.

Try Agai

pfSense.home.arpa Problem loading pa Problem loading pa Problem loading pa Problem loading pa

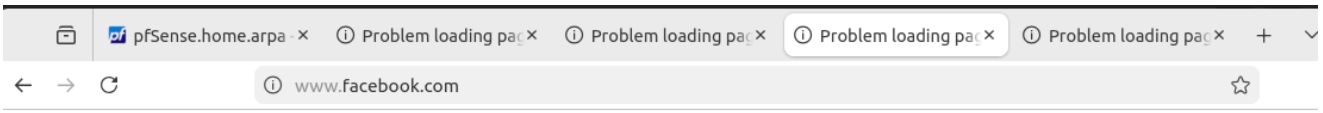
youtube.com

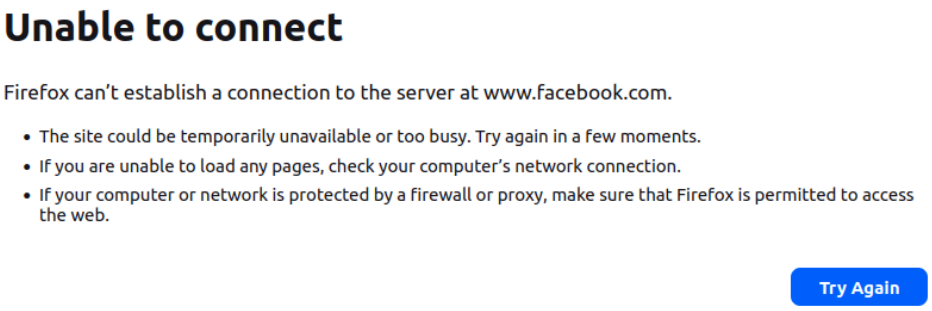
Unable to connect

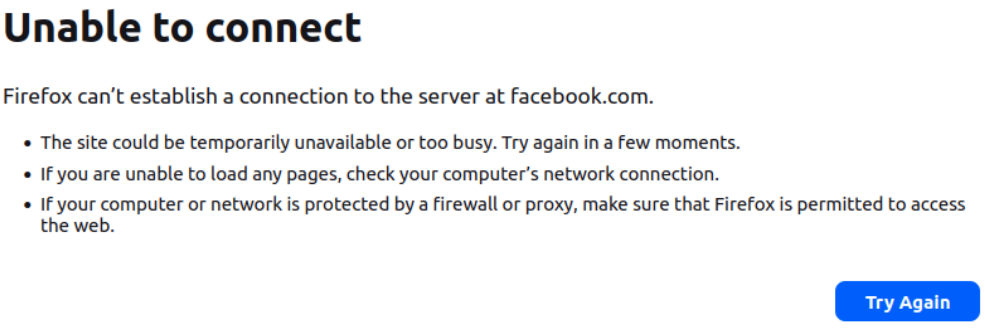
Firefox can't establish a connection to the server at youtube.com.

- The site could be temporarily unavailable or too busy. Try again in a few moments.
- If you are unable to load any pages, check your computer's network connection.
- If your computer or network is protected by a firewall or proxy, make sure that Firefox is permitted to access the web.

Try Again



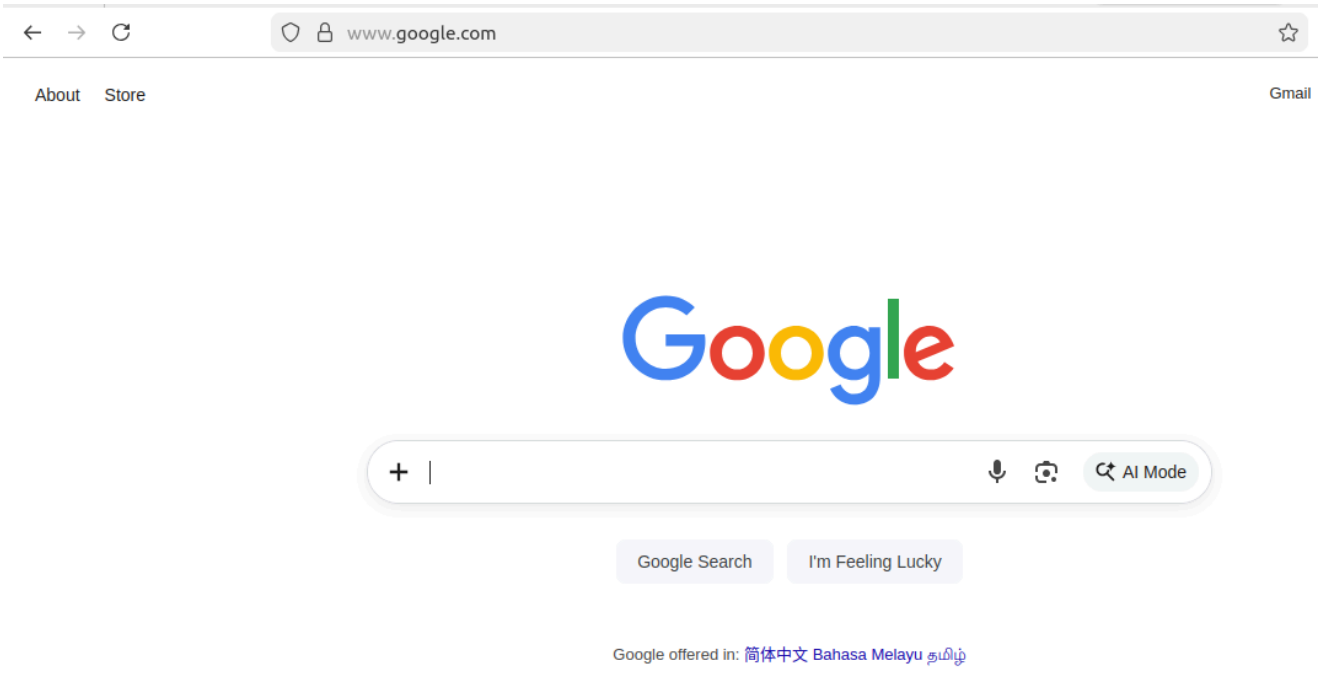




Try:

- `https://www.google.com` - Should work

Result: ✅ YouTube & Facebook blocked via DNS, other sites work



Summary

Final Firewall Rules

→ Firewall → Rules → LAN

[📸 Screenshot: Complete firewall rules list showing all 3 rules in order]

Rules configured:

#	Rule	Protocol	Port	Source	Destination	Action
1	Block ping to VM B	ICMP	-	LAN net	192.168.35.145	Block
2	Block HTTP	TCP	80	LAN net	any	Block
3	Block Telnet	TCP	23	LAN net	any	Block
4	Block YouTube/Facebook	DNS Override	-	-	6 domains	0.0.0.0

Verification Summary

Test	Source	Destination	Expected	Status
Ping VM B	VM A (192.168.132.101)	192.168.35.145	Blocked	✅
Ping pfSense	VM A	192.168.132.250	Success	✅
Ping Internet	VM A	8.8.8.8	Success	✅
HTTP access	VM A	any	Blocked	✅
HTTPS access	VM A	any	Success	✅
Telnet	VM A	VM B	Blocked	✅
SSH	VM A	VM B	Success	✅
YouTube DNS	VM A	youtube.com	0.0.0.0	✅
Facebook DNS	VM A	facebook.com	0.0.0.0	✅
Google DNS	VM A	google.com	Real IP	✅

Explanation

Rule 1: Block ICMP to VM B

How it works:

- **Match**: ICMP protocol, source = LAN (192.168.132.0/24), destination = 192.168.35.145
- **Action**: Drop packet silently
- **Result**: VM A cannot ping VM B specifically

Why specific IP?

- Only blocks ping to VM B
- pfSense gateway still pingable (192.168.132.250)
- Internet still pingable (8.8.8.8)

Rule 2: Block HTTP (Port 80)

How it works:

- **Match**: TCP protocol, destination port = 80, source = LAN
- **Action**: Block all HTTP traffic
- **Result**: Cannot access any HTTP website

Why HTTP but not HTTPS?

- HTTP (port 80) = No encryption → Insecure
- HTTPS (port 443) = Encrypted → Secure
- Force users to use secure connections only

Rule 3: Block Telnet (Port 23)

How it works:

- **Match**: TCP protocol, destination port = 23
- **Action**: Block Telnet connections
- **Result**: Cannot use Telnet to connect to any server

Why block Telnet?

- Telnet transmits passwords in **plain text**
- Easy for hackers to intercept
- SSH (port 22) is encrypted alternative
- Security best practice: Block Telnet, use SSH

Rule 4: DNS Override for YouTube/Facebook

How it works:

1. VM A queries DNS: "What is IP of youtube.com?"
2. pfSense DNS Resolver checks Host Overrides
3. Finds match: youtube.com → 0.0.0.0
4. Returns 0.0.0.0 to VM A
5. VM A tries to connect to 0.0.0.0 → Fails

Why not block by IP?

- YouTube/Facebook use CDNs (Content Delivery Networks)
- Have 100+ different IPs globally
- IPs change frequently
- Impossible to maintain IP block list

Wildcard (*) usage:

- `*.youtube.com` matches:
  -
  - m.youtube.com

    - music.youtube.com
    - All subdomains

Limitations:

- Users can bypass by using external DNS (8.8.8.8, 1.1.1.1)
- VPN can bypass
- Need to force all DNS through pfSense

Firewall Rule Processing

Order matters:

1. Rules processed **top to bottom**
2. **First match wins**
3. Processing stops at first match
4. More specific rules should be placed higher

Example:

```
Rule 1: Block ICMP to 192.168.35.145  (specific)
Rule 2: Block HTTP port 80            (general)
Rule 3: Block Telnet port 23          (general)
...
Rule X: Allow LAN to any              (implicit default)
```

If packet matches Rule 1 → Block, stop processing
If packet doesn't match Rule 1 → Check Rule 2
Continue until match or reach default rule

Conclusion

What We Accomplished

1. ✅ Installed pfSense firewall
2. ✅ Configured WAN and LAN interfaces
3. ✅ Set up DHCP server
4. ✅ Created 4 firewall rules:
    - ICMP blocking (Layer 3)
    - Port-based blocking (Layer 4)

    - DNS-based blocking (Layer 7)
5. ✅ Tested and verified all rules

Skills Learned

Technical Skills:

- pfSense installation and configuration
- Network interface management
- Firewall rule creation
- Protocol understanding (ICMP, TCP, DNS)
- Network testing and troubleshooting

Security Concepts:

- Packet filtering
- Stateful firewall
- Defense in depth
- DNS-based content filtering
- Security vs usability trade-offs

Real-world Applications

Corporate Networks:

- Block insecure protocols (Telnet, HTTP)
- Control employee internet access
- Prevent data leaks

Home Networks:

- Parental controls
- Block malicious sites
- Protect IoT devices

Lab Environment:

- Learn security concepts safely
- Test configurations
- Understand attack/defense

Screenshots Required

Total screenshots needed: **~25-30**

Setup Phase:

1. pfSense Install menu
2. Installation progress
3. Interface assignment (WAN=em0, LAN=em1)
4. LAN IP configuration complete
5. VM A IP from DHCP (192.168.132.101)
6. Ping to pfSense successful
7. VM B IP configuration (192.168.35.145)
8. VM B services listening
9. pfSense login page
10. pfSense Dashboard

Before Blocking:
11. VM A ping VM B success
12. VM A telnet VM B success
13. HTTP website loading

Task 1.1:
14. ICMP rule configuration
15. ICMP rule in rules list
16. Failed ping to VM B
17. Successful ping to pfSense

Task 1.2:
18. HTTP rule configuration
19. HTTP website blocked

Task 1.3:
20. Telnet rule configuration
21. Telnet connection failed

Task 1.4:
22. DNS Resolver with 6 overrides
23. nslookup showing 0.0.0.0
24. YouTube blocked in browser
25. Facebook blocked in browser

Summary:
26. Complete rules list (all 3 rules)

Time Estimate

Total time: ~1.5 hours

- pfSense VM creation: 10 min
- pfSense installation: 15 min
- Interface configuration: 10 min
- VM A/B configuration: 15 min
- Task 1 (4 rules): 30 min
- Testing & screenshots: 20 min

Report Format

File name: `[NT140.PXX.ANTT.X]-Lab5_MSSV1_MSSV2.pdf`

Structure:

1. Cover page (Lab info, student info)
2. Network topology
3. Environment setup (brief)
4. Task 1 implementation (main content)
    - Each rule: config + screenshot + verification
5. Summary table
6. Explanation
7. Conclusion

Tips:

- Focus on results, not theory
- Screenshot every verification
- Brief explanations
- Clear formatting

Lab 5 Complete! ✅

For educational purposes only - NT140 Network Security Course